

FOSD — Unit 8

I/O & Storage

GATE CSE PYQ source-page collection • 2017–2026

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- The question pages below are taken directly from the original source PDFs supplied in this conversation.
- Original question wording, options, tables, code, and figures are preserved on those pages.
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- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
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Included source pages

CS2018 — source page 6 — selected Q9
CS2018 — source page 22 — selected Q53
CS2020 — source page 4 — selected Q3
CS2020 — source page 12 — selected Q35
CS2022 — source page 14 — selected Q17
CS2023 — source page 32 — selected Q34
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Q.9 The following are some events that occur after a device controller issues an interrupt while process L is under execution.

- (P) The processor pushes the process status of L onto the control stack.
- (Q) The processor finishes the execution of the current instruction.
- (R) The processor executes the interrupt service routine.
- (S) The processor pops the process status of L from the control stack.
- (T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

- (A) QPTRS (B) PTRSQ (C) TRPQS (D) QTPRS

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- (A) $(D - M) / (X - M)$ (B) $(X - M) / (D - M)$
(C) $(D - X) / (D - M)$ (D) $(X - M) / (D - X)$

Q.11 In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set $E1$ to entity set $E2$. Assume that $E1$ and $E2$ participate totally in R and that the cardinality of $E1$ is greater than the cardinality of $E2$.

Which one of the following is true about R ?

- (A) Every entity in $E1$ is associated with exactly one entity in $E2$.
- (B) Some entity in $E1$ is associated with more than one entity in $E2$.
- (C) Every entity in $E2$ is associated with exactly one entity in $E1$.
- (D) Every entity in $E2$ is associated with at most one entity in $E1$.

- Q.53 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

- Q.54 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

For parameters a and b , both of which are $\omega(1)$, $T(n) = T(n^{1/a}) + 1$, and $T(b) = 1$.

Then $T(n)$ is

- (A) $\Theta(\log_a \log_b n)$
- (B) $\Theta(\log_{ab} n)$
- (C) $\Theta(\log_b \log_a n)$
- (D) $\Theta(\log_2 \log_2 n)$

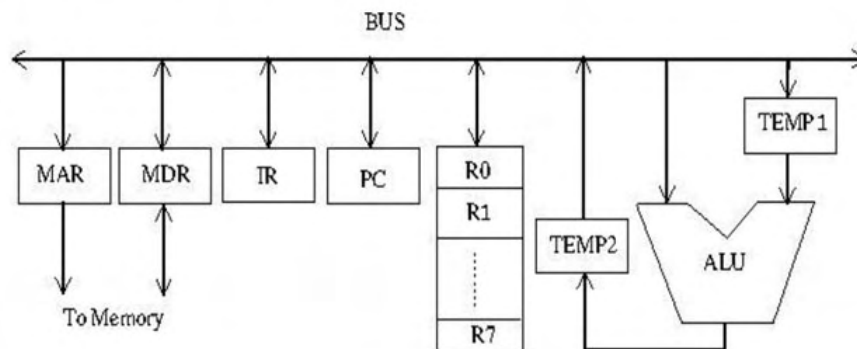
Q.No. 3 Consider the following statements.

- I. Daisy chaining is used to assign priorities in attending interrupts.
- II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt.
- III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.
- IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.

Which of the above statements is/are TRUE?

- (A) I and II only
- (B) I and IV only
- (C) I and III only
- (D) III only

Q.No. 4 Consider the following data path diagram.



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r$, $TEMP1_r$, ALU_{add} , $TEMP2_w$
2. $R1_r$, $TEMP1_w$
3. PC_r , MAR_w , MEM_r
4. $TEMP2_r$, $R0_w$
5. MDR_r , IR_w

Which one of the following is the correct order of execution of the above steps?

- (A) 2, 1, 4, 5, 3
- (B) 1, 2, 4, 3, 5
- (C) 3, 5, 2, 1, 4
- (D) 3, 5, 1, 2, 4

Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time.

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests.

Which one of the following statements is FALSE?

- (A) T is serviced before P.
- (B) Q is serviced after S, but before T.
- (C) The head reverses its direction of movement between servicing of Q and P.
- (D) R is serviced before P.

Q.No. 36 Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?

- (A) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a prime attribute.
- (B) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.
- (C) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key.
- (D) A cell in R holds a set instead of an atomic value.

Q.No. 37 Consider a schedule of transactions T_1 and T_2 :

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for “Read(X)” and WX stands for “Write(X)”. Which one of the following schedules is conflict equivalent to the above schedule?

- (A)

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit
- (B)

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit
- (C)

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit
- (D)

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

Q.No. 38

GATE 2022 Computer Science and Information Technology (CS)

Q.17	Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?
(A)	DMA based I/O transfer
(B)	Interrupt driven I/O transfer
(C)	Polling based I/O transfer
(D)	Programmed I/O transfer
Q.18	Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1+R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?
(A)	$R1 = 1011$ and $R2 = 1110$
(B)	$R1 = 1100$ and $R2 = 1010$
(C)	$R1 = 0011$ and $R2 = 0100$
(D)	$R1 = 1001$ and $R2 = 1111$

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Q.34	A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes $100\ \mu\text{s}$ (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional $200\ \mu\text{s}$ to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke. In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke. The ratio $\frac{T_1}{T_2}$ is _____. (Rounded off to one decimal place)

- Q.53 Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located *randomly* on the disk. Assuming average rotational latency, the total time (*in seconds, rounded off to 2 decimal places*) to read the entire file from the disk is _____
- Q.54 Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (*in MSS*) during the third RTT will be _____
- Q.55 Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (*in bits*) required for collision detection is _____
- Q.56 A functional dependency $F: X \rightarrow Y$ is termed as a *useful functional dependency* if and only if it satisfies all the following three conditions:
- X is not the empty set.
 - Y is not the empty set.
 - Intersection of X and Y is the empty set.
- For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____
- Q.57 A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____

Q.53 Consider two set-associative cache memory architectures: **WBC**, which uses the write back policy, and **WTC**, which uses the write through policy. Both of them use the LRU (*Least Recently Used*) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

- (A) A read miss in **WBC** never evicts a dirty block
- (B) A read miss in **WTC** never triggers a write back operation of a cache block to main memory
- (C) A write hit in **WBC** can modify the value of the dirty bit of a cache block
- (D) A write miss in **WTC** always writes the victim cache block to main memory before loading the missed block to the cache

Q.54 Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

Q.55 The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (*ns*). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 *ns* when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is _____

Q.56 A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (*rounded off to two decimal places*) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

Q.18	Consider the following two statements about interrupt handling mechanisms in a CPU. S1: In non-vectorized interrupt mechanism, it usually takes more time to start the Interrupt Service Routine (ISR) when compared to that in a vectored interrupt mechanism. S2: In daisy-chain interrupt mechanism, the CPU polls all the input devices individually to determine the source of the interrupt. Which one of the following options is correct with respect to S1 and S2 ?
(A)	Both S1 and S2 are true
(B)	Both S1 and S2 are false
(C)	S1 is true and S2 is false
(D)	S1 is false and S2 is true

Q.57	Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (answer in integer)																											
Q.58	Let X be a random variable which takes values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Further, $\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 5) = \Pr(X = 7) = \frac{1}{6}$ and $\Pr(X = 3) = \Pr(X = 4) = \Pr(X = 6) = \Pr(X = 8) = \frac{1}{12}$. The expected value of X, denoted by $E[X]$, is equal to _____. (rounded off to two decimal places)																											
Q.59	Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400. The sector size is 1024 bytes. It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: 5, 12 and 7. The total time for the data transfer (in milliseconds) from the hard disk is _____. (rounded off to one decimal place)																											
Q.60	The EX stage of a pipelined processor performs the memory read operations for LOAD instructions, and the operations for the arithmetic and logic instructions. Let t_{EX} denote the time taken by the EX stage to perform the operation for an instruction. For each instruction type, the values of t_{EX} and M (the number of instructions of that type in a sequence of 100 instructions for a program P), are given in the table below. The duration of the pipeline clock cycle is 1 nanosecond. Assume that the latch time for the interstage buffers in the pipeline is negligible. <table><tr><th>Instruction</th><th>t_{EX} in nanoseconds</th><th>M</th></tr><tr><td>LOAD</td><td>1.8</td><td>15</td></tr><tr><td>IMUL</td><td>1.5</td><td>10</td></tr><tr><td>IDIV</td><td>2.5</td><td>5</td></tr><tr><td>FADD</td><td>1.7</td><td>10</td></tr><tr><td>FSUB</td><td>1.7</td><td>5</td></tr><tr><td>FMUL</td><td>2.8</td><td>15</td></tr><tr><td>FDIV</td><td>3.2</td><td>5</td></tr><tr><td>All other instructions</td><td>Less than 1.0</td><td>35</td></tr></table> When program P is executed, the number of clock cycles for which the pipeline is stalled due to structural hazards in the EX stage is _____. (answer in integer)	Instruction	t_{EX} in nanoseconds	M	LOAD	1.8	15	IMUL	1.5	10	IDIV	2.5	5	FADD	1.7	10	FSUB	1.7	5	FMUL	2.8	15	FDIV	3.2	5	All other instructions	Less than 1.0	35
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